

3ª Lista de Exercícios E.D.O.

1. Verifique se a série geométrica é convergente ou divergente. Em caso de convergência calcule sua soma:

(a) $10 - 2 + 0,4 - 0,08 + \dots$ (b) $\sum_{n=1}^{\infty} 6(0,9)^{n-1}$ (c) $\sum_{n=0}^{\infty} \frac{(10)^n}{(-9)^{n-1}}$

2. Verifique se a série é convergente ou divergente.

(a) $\sum_{n=1}^{\infty} \ln\left(\frac{n}{2n+5}\right)$ (b) $\sum_{n=1}^{\infty} 5\left(\frac{2}{3}\right)^{n-1}$ (c) $\sum_{n=1}^{\infty} \frac{(-6)^{n-1}}{5^{n-1}}$
(d) $\sum_{n=1}^{\infty} 3^{-n} 8^{n+1}$ (e) $\sum_{n=1}^{\infty} \frac{4^{n+1}}{5^n}$ (f) $\sum_{n=1}^{\infty} \frac{n}{n+5}$
(g) $\sum_{n=1}^{\infty} \left(\frac{3}{n}\right)$ (h) $\sum_{n=1}^{\infty} \frac{1}{n(n+2)}$ (i) $\sum_{n=1}^{\infty} \frac{3^n + 2^n}{6^n}$
(j) $\sum_{n=1}^{\infty} \frac{2}{n^2 + 4n + 3}$ (l) $\sum_{n=1}^{\infty} \frac{n}{\sqrt{1+n^2}}$ (m) $\sum_{n=1}^{\infty} \left(\frac{1}{2^{n-1}} + \frac{2}{3^{n-1}}\right)$

3. Determinar os valores de x para os quais a série converge:

(a) $\sum_{n=1}^{\infty} \frac{x^n}{3^n}$ (b) $\sum_{n=1}^{\infty} (x-4)^n$ (c) $\sum_{n=1}^{\infty} 4^n x^n$
(d) $\sum_{n=1}^{\infty} \frac{(x+3)^n}{2^n}$ (e) $\sum_{n=1}^{\infty} \frac{1}{x^n}$

4. Teste a série para convergência ou divergência.

(a) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\sqrt{n}}$ (b) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\sqrt{3n-1}}$ (c) $\sum_{n=1}^{\infty} (-1)^n \frac{2n}{4n+1}$
(d) $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{\sqrt{n}}{n+4}$ (e) $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{n}{2^n}$

5. Determine $x > 0$ para que a série seja convergente.

(a) $\sum_{n=1}^{\infty} \frac{x^n}{n^2}$ (b) $\sum_{n=1}^{\infty} \frac{x^n}{2^n}$ (c) $\sum_{n=1}^{\infty} \frac{x^n}{1 \cdot 3 \cdot 5 \dots (2n+1)}$
(d) $\sum_{n=1}^{\infty} \frac{x^n}{\ln n}$ (e) $\sum_{n=1}^{\infty} \frac{x^n}{n^n}$ (f) $\sum_{n=1}^{\infty} \frac{n! x^n}{n^n}$

6. Use o teste da Raiz ou da Razão para verificar se a série converge ou diverge.

(a) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n\sqrt{n}}$ (b) $\sum_{n=1}^{\infty} \frac{n(-3)^n}{4^{n-1}}$ (c) $\sum_{n=1}^{\infty} \left(\frac{3n}{1+8n}\right)^n$
(d) $\sum_{n=1}^{\infty} \frac{(-2)^{2n}}{n^n}$ (e) $\sum_{n=1}^{\infty} \frac{10^n}{(n+1)4^{2n+1}}$ (f) $\sum_{n=1}^{\infty} \frac{(2n)^n}{n^{2n}}$

Lista 2 - EDO

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1-

a) $10 - 2 + 0,4 - 0,08 + \dots$

$$r = -\frac{1}{5} < 1 \quad \therefore \text{converge}$$

$$\sum_{n=0}^{\infty} 10 \cdot \left(-\frac{1}{5}\right)^n = \frac{10}{1 - (-1/5)} = 10 \cdot \frac{5}{4} = \frac{50}{4} //$$

b) $\sum_{n=1}^{\infty} 6(0,9)^{n-1}$

$$r = 0,9 < 1 \quad \therefore \text{converge}$$

$$\sum_{n=1}^{\infty} 6(0,9)^{n-1} = \frac{6}{1 - 0,9} = \frac{6}{0,1} = 6 \cdot 10 = 60 //$$

c) $\sum_{n=0}^{\infty} \frac{(10)^n}{(-9)^{n-1}} = -9 + 10 - \frac{10^2}{9} + \frac{10^3}{(-9)^2}$

$$r = \left| \frac{-10}{9} \right| = \frac{10}{9} > 1 \quad \therefore \text{diverge} //$$

2-

$$a) \sum_{n=1}^{\infty} \ln \left(\frac{n}{2n+5} \right)$$

$$\lim_{n \rightarrow \infty} \ln \left(\frac{n}{2n+5} \right) = \ln \left[\lim_{n \rightarrow \infty} \frac{n}{2n+5} \right] \stackrel{L.H}{=} \ln \frac{1}{2} \neq 0$$

$\therefore$ diverge

$$b) \sum_{n=1}^{\infty} 5 \left(\frac{2}{3} \right)^{n-1}$$

$$r = \frac{2}{3} < 1 \quad \therefore \text{converge}$$

$$c) \sum_{n=0}^{\infty} \frac{(-6)^{n-1}}{5^{n-1}} = \sum_{n=0}^{\infty} \left(\frac{-6}{5} \right)^{n-1}$$

$$r = \left| \frac{-6}{5} \right| = \frac{6}{5} > 1 \quad \therefore \text{diverge}$$

$$d) \sum_{n=1}^{\infty} 3^{-n} 8^{n+1} = \sum_{n=1}^{\infty} 8 \cdot \left(\frac{8}{3} \right)^n$$

$$r = \frac{8}{3} > 1 \quad \therefore \text{diverge}$$

$$e) \sum_{n=1}^{\infty} \frac{4^{n+1}}{5^n} = \sum_{n=1}^{\infty} 4 \left(\frac{4}{5} \right)^n$$

$$r = \frac{4}{5} < 1 \quad \therefore \text{converge}$$

$$f) \sum_{n=1}^{\infty} \frac{n}{n+5}$$

$$\lim_{n \rightarrow \infty} \frac{n}{n+5} \stackrel{L.H}{=} \lim_{n \rightarrow \infty} \frac{1}{5} = \frac{1}{5} \neq 0 \therefore \text{diverge}$$

$$g) \sum_{n=1}^{\infty} \left(\frac{3}{n} \right) = 3 \sum_{n=1}^{\infty} \left(\frac{1}{n} \right)$$

série harmônica, portanto diverge

$$h) \sum_{n=1}^{\infty} \frac{1}{n(n+2)} = \sum_{n=1}^{\infty} \frac{1}{n^2+2n}$$

comparando com $\frac{1}{n^2}$:

$$\lim_{n \rightarrow \infty} \frac{\frac{1}{n^2+2n}}{\frac{1}{n^2}} = \lim_{n \rightarrow \infty} \frac{n^2}{n^2+2n} \stackrel{L.H}{=} \lim_{n \rightarrow \infty} \frac{2n}{2n+2} \stackrel{L.H}{=} \lim_{n \rightarrow \infty} \frac{2}{2} = 1 > 0$$

$\therefore$ a série $\sum_{n=1}^{\infty} \frac{1}{n(n+2)}$ também converge.

$$i) \sum_{n=1}^{\infty} \frac{3^n + 2^n}{6^n} = \sum_{n=1}^{\infty} \left(\frac{1}{2} \right)^n_{\text{I}} + \sum_{n=1}^{\infty} \left(\frac{1}{3} \right)^n_{\text{II}}$$

$$\text{I} \quad r = \frac{1}{2} < 1 \therefore \text{converge}$$

$$\text{II} \quad r = \frac{1}{3} < 1 \therefore \text{converge}$$

Logo $\sum_{n=1}^{\infty} \frac{3^n + 2^n}{6^n}$ também converge.

$$j) \sum_{n=1}^{\infty} \frac{2}{n^2+4n+3}$$

comparando con $\frac{1}{n^2}$:

$$\lim_{n \rightarrow \infty} \frac{\frac{2}{n^2+4n+3}}{\frac{1}{n^2}} = \lim_{n \rightarrow \infty} \frac{2n^2}{n^2+4n+3} \stackrel{L.H}{=} \lim_{n \rightarrow \infty} \frac{4n}{2n+4} \stackrel{L.H}{=} \lim_{n \rightarrow \infty} \frac{4}{2} = 2 > 0 \quad \therefore \sum_{n=1}^{\infty} \frac{2}{n^2+4n+3} \text{ también converge.}$$

$$l) \sum_{n=1}^{\infty} \frac{n}{\sqrt{1+n^2}}$$

$$\lim_{n \rightarrow \infty} \frac{n}{\sqrt{1+n^2}} = \lim_{n \rightarrow \infty} \frac{\cancel{n}}{\cancel{n} \sqrt{\cancel{n}^2+1}} = 1 \neq 0 \quad \therefore \text{diverge}$$

$$m) \sum_{n=1}^{\infty} \left(\frac{1}{2^{n-1}} + \frac{2}{3^{n-1}} \right) = \sum_{n=1}^{\infty} \frac{1}{2^{n-1}} \text{ I} + \sum_{n=1}^{\infty} \frac{1}{3^{n-1}} \text{ II}$$

$$\text{I} \quad r = \frac{1}{2} < 1 \quad \therefore \text{converge}$$

$$\text{II} \quad r = \frac{1}{3} < 1 \quad \therefore \text{converge}$$

$$\text{logo, } \sum_{n=1}^{\infty} \left(\frac{1}{2^{n-1}} + \frac{2}{3^{n-1}} \right) \text{ también converge}$$

3-

$$a) \sum_{n=1}^{\infty} \frac{x^n}{3^n} = \sum_{n=1}^{\infty} \left(\frac{x}{3} \right)^n$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{x}{3} \right)^n \right|} = \lim_{n \rightarrow \infty} \left| \frac{x}{3} \right| = x \lim_{n \rightarrow \infty} \left| \frac{1}{3} \right| = \left| \frac{x}{3} \right|$$

converge com $L < 1$.

$$\left| \frac{x}{3} \right| < 1 \rightarrow |x| < 3 \rightarrow -3 < x < 3$$

$$\text{Se } x = -3: \sum_{n=1}^{\infty} \frac{(-3)^n}{3^n} = \sum_{n=1}^{\infty} (-1)^n \text{ diverge}$$

$$\text{Se } x = 3: \sum_{n=1}^{\infty} \frac{3^n}{3^n} = \sum_{n=1}^{\infty} 1 \text{ diverge}$$

$$\therefore \sum_{n=1}^{\infty} \frac{x^n}{3^n} \text{ converge para todo } x \in (-3, 3)$$

$$b) \sum_{n=1}^{\infty} (x-4)^n$$

$$r = |x-4|, \text{ converge com } |x-4| < 1$$

$$|x-4| < 1 \rightarrow -1 < x-4 < 1 \rightarrow 3 < x < 5$$

$$\text{Se } x = 3: \sum_{n=1}^{\infty} (-1)^n \text{ diverge}$$

$$\text{Se } x = 5: \sum_{n=1}^{\infty} 5^n \text{ diverge}$$

$$\therefore \sum_{n=1}^{\infty} (x-4)^n \text{ converge para todo } x \in (3, 5)$$

$$c) \sum_{n=1}^{\infty} 4^n x^n = \sum_{n=1}^{\infty} (4x)^n$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{|(4x)^n|} = \lim_{n \rightarrow \infty} |4x| = |4x|$$

$$\text{converge com } |4x| < 1 : -1 < 4x < 1 \rightarrow -\frac{1}{4} < x < \frac{1}{4}$$

$$\text{Se } x = \frac{1}{4} : \sum_{n=1}^{\infty} 1^n = \sum_{n=1}^{\infty} 1 \text{ diverge}$$

$$\text{Se } x = -\frac{1}{4} : \sum_{n=1}^{\infty} (-1)^n \text{ diverge}$$

$$\therefore \sum_{n=1}^{\infty} 4^n x^n \text{ converge para todo } x \in \left(-\frac{1}{4}; \frac{1}{4}\right)$$

$$d) \sum_{n=1}^{\infty} \frac{(x+3)^n}{2^n} = \sum_{n=1}^{\infty} \left(\frac{x+3}{2}\right)^n$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{\left|\left(\frac{x+3}{2}\right)^n\right|} = \lim_{n \rightarrow \infty} \left|\frac{x+3}{2}\right| = \left|\frac{x+3}{2}\right|$$

$$\text{converge com } \left|\frac{x+3}{2}\right| < 1 : |x+3| < 2 \rightarrow -2 < x+3 < 2$$

$$-5 < x < -1$$

$$\text{Se } x = -5 : \sum_{n=1}^{\infty} (-1)^n \text{ diverge}$$

$$\text{Se } x = -1 : \sum_{n=1}^{\infty} 1^n \text{ diverge}$$

$$\therefore \frac{(x+3)^n}{2^n} \text{ converge para todo } x \in (-5; -1)$$

$$e) \sum_{n=1}^{\infty} \frac{1}{x^n} = \sum_{n=1}^{\infty} \left(\frac{1}{x} \right)^n$$

$$r = \left| \frac{1}{x} \right|, \text{ converge com } \left| \frac{1}{x} \right| < 1$$

$$\left| \frac{1}{x} \right| < 1 \rightarrow 1 < |x| \rightarrow x < -1 \text{ ou } x > 1$$

$$\text{se } x = -1: \sum_{n=1}^{\infty} (-1)^n \text{ diverge}$$

$$\text{se } x = 1: \sum_{n=1}^{\infty} 1^n \text{ diverge}$$

$$\therefore \sum_{n=1}^{\infty} \frac{1}{x^n} \text{ converge em } (-\infty; -1) \cup (1; +\infty)$$

4-

$$a) \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\sqrt{n}}$$

$$a_n = \frac{1}{\sqrt{n}}, \text{ é decrescente}$$

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} = 0$$

$\therefore$ a série converge

$$b) \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\sqrt{3n-1}}$$

$$a_n = \frac{1}{\sqrt{3n-1}}, \text{ é decrescente}$$

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{3n-1}} = 0$$

$\therefore$ a série converge.

$$c) \sum_{n=1}^{\infty} (-1)^n \frac{2n}{4n+1}$$

$$a_n = \frac{2n}{4n+1} = \frac{1}{2 + \frac{1}{2n}}, \text{ é decrescente}$$

$$\lim_{n \rightarrow \infty} \frac{2n}{4n+1} \stackrel{L.H}{=} \frac{2}{4} = \frac{1}{2} \neq 0$$

$\therefore$ a série diverge

$$d) \sum_{n=1}^{\infty} (-1)^{n-1} \frac{\sqrt{n}}{n+4}$$

$$a_n = \frac{\sqrt{n}}{n+4}, \text{ é decrescente}$$

$$\lim_{n \rightarrow \infty} \frac{n^{1/2}}{n+4} \stackrel{L.H}{=} \lim_{n \rightarrow \infty} \frac{1}{2\sqrt{n}} = 0$$

$\therefore$ a série é convergente

$$e) \sum_{n=1}^{\infty} (-1)^{n+1} \frac{n}{2^n}$$

$$a_n = \frac{n}{2^n}, \text{ é crescente}$$

$$\lim_{n \rightarrow \infty} \frac{n}{2^n} \stackrel{L.H}{=} \lim_{n \rightarrow \infty} \frac{1}{\cancel{2^n} \ln(n)^{\infty}} = 0$$

$\therefore$ a série converge

5- $x > 0$

$$a) \sum_{n=1}^{\infty} \frac{x^n}{n^2}$$

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{x^{n+1}}{(n+1)^2}}{\frac{x^n}{n^2}} \right| = \lim_{n \rightarrow \infty} \left| \frac{\cancel{x} \cdot x^n}{(n+1)^2} \cdot \frac{n^2}{\cancel{x^n}} \right| = \lim_{n \rightarrow \infty} \left| x \cdot \frac{n^2}{(n+1)^2} \right|$$

$$= |x| \lim_{n \rightarrow \infty} \frac{n^2}{(n+1)^2} \stackrel{L.H}{=} |x| \lim_{n \rightarrow \infty} \frac{\cancel{2n}}{\cancel{2(n+1)}} \stackrel{L.H}{=} |x| \lim_{n \rightarrow \infty} 1 = |x|$$

para convergir $|x| < 1 \rightarrow 0 < x < 1$

se $x = 1$: $\sum_{n=1}^{\infty} \frac{1}{n^2}$, série harmônica $p=2$ $\therefore$ converge

logo, $\sum_{n=1}^{\infty} \frac{x^n}{n^2}$ converge para todo $x \in (0, 1]$

$$b) \sum_{n=1}^{\infty} \frac{x^n}{2^n} = \sum_{n=1}^{\infty} \left(\frac{x}{2} \right)^n$$

$$r = \left| \frac{x}{2} \right|, \text{ converge quando } \left| \frac{x}{2} \right| < 1, \text{ logo } |x| < 2$$

$$0 < x < 2$$

$$\text{se } x = 2: \sum_{n=1}^{\infty} 1, \text{ diverge}$$

$$\therefore \sum_{n=1}^{\infty} \frac{x^n}{2^n} \text{ converge para todo } x \in (0; 2)$$

$$c) \sum_{n=1}^{\infty} \frac{x^n}{1 \cdot 3 \cdot 5 \cdots (2n+1)}$$

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{x^{n+1}}{1 \cdot 3 \cdot 5 \cdots (2n+1)(2n+3)}}{\frac{x^n}{1 \cdot 3 \cdot 5 \cdots (2n+1)}} \right| = \lim_{n \rightarrow \infty} \left| \frac{x^{n+1}}{x^n (2n+3)} \right| = \lim_{n \rightarrow \infty} \left| \frac{x}{2n+3} \right|$$

$$= |x| \lim_{n \rightarrow \infty} \frac{1}{2n+3} = 0 < 1$$

$\therefore$ a série é convergente para qualquer x

$$d) \sum_{n=1}^{\infty} \frac{x^n}{\ln(n)}$$

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{x^{n+1}}{\ln(n+1)}}{\frac{x^n}{\ln(n)}} \right| = |x| \lim_{n \rightarrow \infty} \frac{\ln(n)}{\ln(n+1)} \stackrel{L.H}{=} |x| \lim_{n \rightarrow \infty} \frac{1/2}{1/(n+1)} = |x|$$

para convergir $|x| < 1 \rightarrow 0 < x < 1$

se $x=1$: $\sum_{n=1}^{\infty} \frac{1}{\ln(n)}$, diverge

$\therefore \sum_{n=1}^{\infty} \frac{x^n}{\ln(n)}$ converge para todo $x \in (0; 1)$

$$e) \sum_{n=1}^{\infty} \frac{x^n}{n^n} = \sum_{n=1}^{\infty} \left(\frac{x}{n} \right)^n$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{x}{n} \right)^n \right|} = |x| \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$\therefore \sum_{n=1}^{\infty} \frac{x^n}{n^n}$ converge para todo $x > 0$

$$f) \sum_{n=1}^{\infty} \frac{n! x^n}{n^n}$$

$$\lim_{n \rightarrow \infty} \frac{(n+1) \cdot \cancel{n!} \cdot \cancel{x} \cdot \cancel{x^n}}{\cancel{n!} \cdot \cancel{x^n} \cdot n^n} = \lim_{n \rightarrow \infty} \left| \frac{x \cdot n^n}{(n+1)^n} \right| = |x| \lim_{n \rightarrow \infty} \frac{n^n}{(n+1)^n}$$

$$= |x| \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^n = |x| \lim_{n \rightarrow \infty} \left(\frac{1}{1 + \frac{1}{n}} \right)^n = \frac{|x|}{e}$$

converge se $\frac{|x|}{e} < 1 \rightarrow |x| < e \rightarrow 0 < x < e$

$\therefore \sum_{n=1}^{\infty} \frac{n! x^n}{n^n}$ converge para todo $x \in (0; e)$

6- Usar teste da Raiz ou teste da Razão

$$a) \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n \sqrt{n}} = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^{3/2}}$$

$$a_n = \frac{1}{n^{3/2}}, \text{ decrescente}$$

$$\lim_{n \rightarrow \infty} \frac{1}{n^{3/2}} = 0 \quad \therefore \text{ a série é convergente}$$

$$b) \sum_{n=1}^{\infty} \frac{n(-3)^n}{4^{n-1}}$$

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{(n+1)(-3)^{n+1}}{4^n}}{\frac{n(-3)^n}{4^{n-1}}} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+1)(-3)\cancel{(-3)^n} \cdot \cancel{4^{n-1}}}{4 \cdot \cancel{4^{n-1}} \cdot n \cancel{(-3)^n}} \right| = \lim_{n \rightarrow \infty} \left| \frac{-3n-3}{4n} \right|$$

$$\stackrel{L.H}{=} \left| \frac{-3}{4} \right| < 1 \quad \therefore \text{ converge}$$

$$c) \sum_{n=1}^{\infty} \left(\frac{3n}{1+8n} \right)^n$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{\left| \frac{3n}{1+8n} \right|} = \lim_{n \rightarrow \infty} \frac{3n}{1+8n} \stackrel{L.H}{=} \lim_{n \rightarrow \infty} \frac{3}{8} = \frac{3}{8} < 1$$

$\therefore$ a série converge

$$d) \sum_{n=1}^{\infty} \frac{(-2)^{2n}}{n^n} = \sum_{n=1}^{\infty} \frac{4^n}{n^n}$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{\left| \frac{4}{n} \right|} = \lim_{n \rightarrow \infty} \frac{4}{n} = 0 < 1 \quad \therefore \text{convergente}$$

$$e) \sum_{n=1}^{\infty} \frac{10^n}{(n+1) 4^{2n+1}}$$

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{10^{n+1}}{(n+2) 4^{2(n+1)+1}}}{\frac{10^n}{(n+1) 4^{2n+1}}} \right| = \lim_{n \rightarrow \infty} \left| \frac{10 \cdot \cancel{10^n}}{(n+2) 4^2 \cancel{4^{2n+1}}} \cdot \frac{(n+1) \cancel{4^{2n+1}}}{\cancel{10^n}} \right| = \lim_{n \rightarrow \infty} \frac{10(n+1)}{16(n+2)}$$

$$\stackrel{\text{L.H}}{=} \frac{10}{16} = \frac{5}{8} < 1 \quad \therefore \text{converge}$$

$$f) \sum_{n=1}^{\infty} \frac{(2n)^n}{n^{2n}} = \sum_{n=1}^{\infty} \left(\frac{2n}{n^2} \right)^n$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{2n}{n^2} \right) \right|^n} = \lim_{n \rightarrow \infty} \frac{2n}{n^2} \stackrel{\text{L.H}}{=} \lim_{n \rightarrow \infty} \frac{1}{n} = 0 < 1$$

$\therefore$ a série converge.